

heart of this vision with a focus not just on technological advancement but also on making space technology accessible and affordable for a wider societal impact.

Some of the key milestones that I personally would like to see by 2029 in India are:

1. Dedicated manufacturing clusters, supported by states, with plug-and-play facilities for producing small satellites, payloads, launch vehicle subsystems, ground station equipment and subsystems catering to the global market requirements. The next few years should see India evolving into a significant global supplier, with active participation from Isro's vendors and new space start-ups, by integrating into the global supply chain for space systems.

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2. Currently all GSO communication satellites are either owned by Isro/NSIL or by foreign companies. We would like to see the Indian private sector participate in the huge expansion of the need for communication satellites.

3. An Indian Earth Observation (EO) constellation, being managed and operated by private players.

4. A global data processing and dissemination platform managed by Indian private players.

In the coming years, as the space data requirement increases exponentially, attaining *Atmanirbharta* in space data will be a strategic and commercial imperative. Currently, a large chunk of our space data requirement is being met from foreign satellites, which should gradually shift to Indian sources. And both the EO constellation platform and data processing platform will be an essential step towards *Atmanirbharta* in space data and data products.



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POLICY REFORMS, PRIVATE SECTOR DYNAMISM, AND TECH ADVANCEMENTS WILL MAKE INDIA A GLOBAL SPACE LEADER

5. Complete digital connectivity, even in the remotest parts of the country, made accessible and affordable through satellite broadband. Bringing high-speed internet to the remotest parts of the country through satellites will catalyse socio-economic development, offering opportunities for education, healthcare, and economic change in previously inaccessible regions.

6. India to become a global hub for the launch of small satellites with at least three small launch vehicles owned and operated by private players and the spaceport in Kulasekharapattinam, Tamil Nadu, providing the required launch infrastructure.

7. Downstream applications developed by NGEs with the use of space data and images. These can be to monitor changing climate, extreme weather, use of natural resources, address challenges in agriculture, clean energy, drinking water,

tele-education, telemedicine, and much more, thereby creating new economic opportunities in the country.

Apart from this, Isro will continue to make great strides in the research and development of next-generation space technologies and carrying out interplanetary missions for space exploration, taking Indian presence deep into space from the moon to Venus and more.

The next five years promise to unveil a new era in the Indian space sector, where the focus is not just on technological advancement but also on global collaboration, on leveraging space for socio-economic benefits and on inclusive growth.

The integration of policy reforms, private sector dynamism, and technological advancements will position India not just as a participant but as one of the leaders in shaping the future of global space endeavours! **BT**

The author is Chairman of IN-SPACe. Views are personal